2 In an experiment to determine the specific heat capacity of a liquid, a student heated a fixed mass of the liquid for a fixed duration of time, using an electric heater. The student repeated the experiment three times to find the rise in temperature of the liquid. The following measurements were obtained:

Mass of liquid, m	309 ± 3 g
Voltage applied across heater, V	11.8 ± 0.3 V
Current flow in the heater, I	4.125 ± 0.002 A
Time taken, t	200.0 ± 0.5 s

The rise in temperature θ was recorded for each attempt:

Attempt:	1st	2nd	3rd
θ / K	10.2	9.7	10.5

(a) Estimate the uncertainty in θ.

uncertainty in θ = ... K [1]

(b) Calculate the specific heat capacity c of the liquid.

c = ... J kg^{-1} K^{-1} [2]

(c) Calculate the uncertainty in specific heat capacity *c* of the liquid and express the specific heat capacity *c* together with its uncertainty.

c = ... ± ..... J kg^{-1} K^{-1} [3]

(d) State an assumption made in your calculation of the specific heat capacity *c* of the liquid.

.....

..... [1]